

26.1 What collinearity does to variance

The columns of a model matrix are **collinear** when a nontrivial linear combination of them is zero, and **nearly collinear** when a combination with coefficients of reasonable size is a short vector. The exact case is rank deficiency, settled in [Chapter 8](#). Here $\mathbf{X}$ has full column rank and every linear function of $\boldsymbol{\beta}$ is estimable, yet some are estimated so poorly that in practice they have not been estimated at all; “collinearity” below means this near case.

[Section 19.6](#) wrote the variance of $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ as a sum over the eigenvectors of $\mathbf{X}^\top \mathbf{X}$ (Proposition 19.6.1). This section says which functions collinearity damages, links them to estimability, and separates the statistical problem from the numerical one of [Chapter 10](#).

26.1.1 Canonical coordinates

Throughout the section $\mathbf{X}$ is $n \times p$ of rank p , $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. Write the spectral decomposition (Theorem 1.7.2) as

$$\mathbf{X}^\top \mathbf{X} = \sum_{\ell=1}^p \lambda_\ell \mathbf{v}_\ell \mathbf{v}_\ell^\top, \quad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p > 0,$$

with orthonormal eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_p$, and put $\gamma_\ell = \mathbf{v}_\ell^\top \boldsymbol{\beta}$. The numbers γ_ℓ are the coordinates of $\boldsymbol{\beta}$ in the eigenvector basis, the **canonical coordinates** of the regression. Since $\mathbf{X}\mathbf{v}_\ell$ has squared length λ_ℓ , the mean vector is

$$\mathbf{X}\boldsymbol{\beta} = \sum_{\ell=1}^p \gamma_\ell \mathbf{X}\mathbf{v}_\ell = \sum_{\ell=1}^p \sqrt{\lambda_\ell} \gamma_\ell \mathbf{u}_\ell, \quad \mathbf{u}_\ell = \mathbf{X}\mathbf{v}_\ell / \sqrt{\lambda_\ell}, \quad (26.1.1)$$

and the vectors $\mathbf{u}_\ell$ are the left singular vectors of $\mathbf{X}$ (Theorem 1.8.1). A small λ_ℓ means that a unit change in γ_ℓ moves the mean vector only by $\sqrt{\lambda_\ell}$. The data then carry little information about γ_ℓ , and the following theorem makes this precise.

Theorem 26.1.1. Variances in canonical coordinates

Under the assumptions above, let $\hat{\gamma}_\ell = \mathbf{v}_\ell^\top \hat{\boldsymbol{\beta}}$.

(a) $\hat{\gamma}_\ell = \mathbf{u}_\ell^\top \mathbf{Y} / \sqrt{\lambda_\ell}$, and $\hat{\boldsymbol{\beta}} = \sum_{\ell} \hat{\gamma}_\ell \mathbf{v}_\ell$. The $\hat{\gamma}_\ell$ are uncorrelated, with $\text{Var}(\hat{\gamma}_\ell) = \sigma^2 / \lambda_\ell$.

(b) For every nonzero $\mathbf{a} \in \mathbb{R}^p$,

$$\frac{\sigma^2 \|\mathbf{a}\|^2}{\lambda_1} \leq \text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) \leq \frac{\sigma^2 \|\mathbf{a}\|^2}{\lambda_p},$$

with equality on the left at $\mathbf{a} = \mathbf{v}_1$ and on the right at $\mathbf{a} = \mathbf{v}_p$. The ratio of the largest to the smallest variance of $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ over unit vectors $\mathbf{a}$ is $\kappa(\mathbf{X})^2$.

(c) If $\mathbf{a} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, then $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) \leq \sigma^2 \|\mathbf{a}\|^2 / \lambda_k$: the eigenvalues $\lambda_{k+1}, \dots, \lambda_p$ do not enter. For every $\mathbf{a}$, $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) \geq \sigma^2 (\mathbf{a}^\top \mathbf{v}_p)^2 / \lambda_p$.

(d) $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \sigma^2 \sum_{\ell} 1/\lambda_\ell \geq \sigma^2 / \lambda_p$, while, by Proposition 19.6.1(b), $E\|\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\boldsymbol{\beta}\|^2 = \sigma^2 p$ whatever the eigenvalues.

(e) For each column, $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 \geq \lambda_p$, where $\mathbf{M}_{(j)}$ projects onto the span of the other columns. Hence $\text{Var}(\hat{\beta}_j) \leq \sigma^2 / \lambda_p$.

Proof.

- (a) By Theorem 5.5.1, $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ with covariance $\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 \sum_{\ell} \lambda_{\ell}^{-1} \mathbf{v}_{\ell} \mathbf{v}_{\ell}^\top$. Hence $\mathbf{v}_{\ell}^\top \hat{\beta} = \lambda_{\ell}^{-1} (\mathbf{X} \mathbf{v}_{\ell})^\top \mathbf{Y} = \mathbf{u}_{\ell}^\top \mathbf{Y} / \sqrt{\lambda_{\ell}}$, and $\text{Cov}(\mathbf{v}_{\ell}^\top \hat{\beta}, \mathbf{v}_m^\top \hat{\beta}) = \sigma^2 \mathbf{v}_{\ell}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{v}_m$, which is $\sigma^2 / \lambda_{\ell}$ if $\ell = m$ and zero otherwise. The expansion of $\hat{\beta}$ holds because the $\mathbf{v}_{\ell}$ form an orthonormal basis.
- (b) Write $c_{\ell} = \mathbf{a}^\top \mathbf{v}_{\ell}$, so that $\sum_{\ell} c_{\ell}^2 = \|\mathbf{a}\|^2$. By Proposition 19.6.1(a), $\text{Var}(\mathbf{a}^\top \hat{\beta}) = \sigma^2 \sum_{\ell} c_{\ell}^2 / \lambda_{\ell}$, which lies between $\sigma^2 \|\mathbf{a}\|^2 / \lambda_1$ and $\sigma^2 \|\mathbf{a}\|^2 / \lambda_p$, with the stated equality cases. The singular values of $\mathbf{X}$ are $\sqrt{\lambda_{\ell}}$ (Theorem 1.8.1), so the ratio of the extremes is $\lambda_1 / \lambda_p = \kappa(\mathbf{X})^2$ by Definition 1.8.1.
- (c) If $\mathbf{a} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, then $c_{\ell} = 0$ for $\ell > k$ and $\sum_{\ell \leq k} c_{\ell}^2 / \lambda_{\ell} \leq \|\mathbf{a}\|^2 / \lambda_k$. The lower bound keeps only the term $\ell = p$.
- (d) $E\|\hat{\beta} - \beta\|^2 = \text{tr Cov}(\hat{\beta}) = \sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$, and the eigenvalues of $(\mathbf{X}^\top \mathbf{X})^{-1}$ are the $1/\lambda_{\ell}$. The statement about fitted values is Proposition 19.6.1(b) multiplied by n .
- (e) $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j = \mathbf{x}_j - \mathbf{X}_{(j)}\mathbf{c}$ for some $\mathbf{c}$, which is $\mathbf{X}\mathbf{b}$ for a vector $\mathbf{b}$ with $b_j = 1$. By Theorem 1.7.7, $\|\mathbf{X}\mathbf{b}\|^2 \geq \lambda_p \|\mathbf{b}\|^2 \geq \lambda_p$. The variance bound follows from Proposition 7.2.4. ■

Part (b) is the precise sense in which $\kappa(\mathbf{X})^2$ measures collinearity: it is the ratio of the worst to the best precision among linear functions of the same length. Part (e) is the Frisch–Waugh–Lovell picture of Theorem 6.6.2: a coefficient is imprecise when its column's part orthogonal to the others is short, and no such part is shorter than $\sqrt{\lambda_p}$.

Idea. Collinearity does not make least squares imprecise in general. It makes it imprecise in a few directions of coefficient space, those of the eigenvectors with small eigenvalues, and precise in the others. Whether it matters depends on whether the question asked points along a weak direction.

Example. Two regressors that move together

Draw $n = 30$ values x_{i1} from a standard normal distribution and set $x_{i2} = x_{i1} + 0.15z_i$ with independent standard normal z_i . The sample correlation is 0.9951. Centre both regressors and fit $E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, so that the intercept is orthogonal to the slopes and the slope block of $\mathbf{X}^\top \mathbf{X}$ carries all the collinearity. Its eigenvalues are 70.05 and 0.1718. The weak eigenvector is close to $(1, -1)^\top / \sqrt{2}$ (its first coordinate is 0.718 in absolute value), so the weak canonical coordinate is essentially the difference $\beta_1 - \beta_2$. In units of σ , the standard deviations are

Function	Standard deviation
$\hat{\beta}_1$	1.734
$\hat{\beta}_2$	1.682
$\hat{\beta}_1 + \hat{\beta}_2$	0.177
$\hat{\beta}_1 - \hat{\beta}_2$	3.412

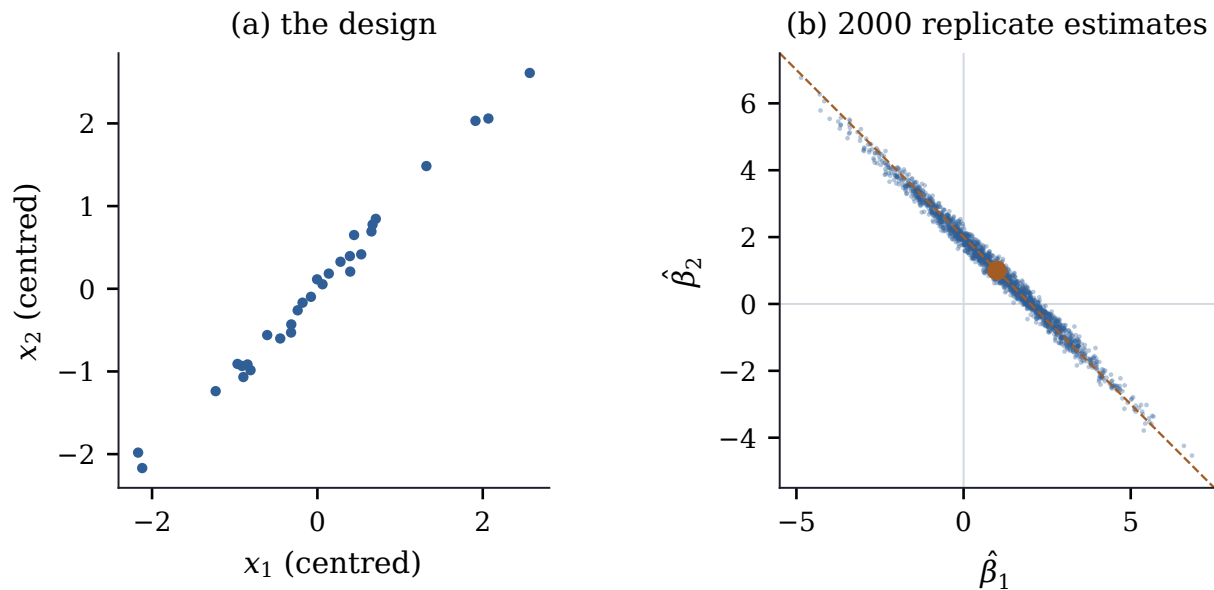


Figure 26.1.1: (a) The design of [Example \(Two regressors that move together\)](#). (b) Estimates $(\hat{\beta}_1, \hat{\beta}_2)$ from 2000 replicate responses, true value marked: long in the weak direction $(1, -1)$, thin along $(1, 1)$; dashed, $\beta_1 + \beta_2 = 2$.

Had x_1 been orthogonal to x_2 with the same spread, the first entry would have been 0.171, smaller by the factor $10.1 = \sqrt{102.4}$, the root of the variance inflation factor of Section 26.2. The sum of the slopes is estimated about as well as a slope in an orthogonal design, the difference worst of all. The ratio of the largest to the smallest variance over unit vectors is 407.8, the squared condition number of the centred slope columns ([Theorem 26.1.1\(b\)](#)).

[Figure 26.1.1](#) shows what this means for a single data set. With the design fixed and true slopes $\beta_1 = \beta_2 = 1$, the estimates from 2000 replicate responses lie along the line $\beta_1 + \beta_2 = 2$. The estimate of β_2 ranges from -4.535 to 7.359 and is negative in 26.1 per cent of the replicates, while $\hat{\beta}_1 + \hat{\beta}_2$ stays between 1.440 and 2.639. Every data set says clearly that the two regressors together matter, yet a quarter of them give β_2 the wrong sign, and 54.1 per cent give at least one of the two slopes the wrong sign.

```
import numpy as np
rng = np.random.default_rng(2601)
n, sigma = 30, 1.0
x1 = rng.normal(size=n)
x2 = x1 + 0.15 * rng.normal(size=n)           # x2 nearly repeats x1
x1c, x2c = x1 - x1.mean(), x2 - x2.mean()     # centring makes the intercept
                                                orthogonal
X = np.column_stack([np.ones(n), x1c, x2c])
G = np.linalg.inv(X.T @ X)                     # Cov(beta_hat) / sigma^2

lam, V = np.linalg.eigh(X[:, 1:].T @ X[:, 1:]) # canonical coordinates of the
                                                slopes
print("correlation of x1 and x2:", round(np.corrcoef(x1, x2)[0, 1], 4))
print("eigenvalues:", np.round(lam, 3), " weak direction:", np.round(V[:, 0], 3))

def sd(a):
    """Standard deviation of a' beta_hat, in units of sigma."""
    return np.sqrt(a @ G @ a)
```

```
print("sd of b1, b2   :", round(sd(np.array([0, 1, 0])), 3), round(sd(np.array([0,
    0, 1])), 3))
print("sd of b1 + b2   :", round(sd(np.array([0, 1, 1])), 3))
print("sd of b1 - b2   :", round(sd(np.array([0, 1, -1])), 3))
```

[Run this code online](#)

```
beta = np.array([0.0, 1.0, 1.0])          # equal true slopes
reps = 2000
Y = X @ beta + sigma * rng.normal(size=(reps, n))  # 2000 replicate responses,
    same design
B = np.linalg.solve(X.T @ X, X.T @ Y.T).T          # one row of estimates per
    replicate
print("share of replicates with b2 < 0:", np.mean(B[:, 2] < 0))
print("share with b1 < 0 or b2 < 0:", np.mean((B[:, 1:] < 0).any(axis=1)))
print("range of b1 + b2:", np.round([B[:, 1:].sum(1).min(), B[:, 1:].sum(1).max()],
    3))
```

[Run this code online](#)

26.1.2 The symptoms

The classical symptoms all follow from [Theorem 26.1.1](#): large standard errors for single coefficients alongside small ones for some combinations; coefficients of implausible size or sign; instability under deletion of a few cases; and a significant F test for a group in which no single t test is significant, because the group statistic is governed by the best combination of its members ([Proposition 11.6.2](#)). None of them means the model is wrong.

26.1.3 Near-estimability

When the near-dependence is a small perturbation of an exact one, the functions that stay precise are exactly those estimable in the limiting, rank-deficient design.

Proposition 26.1.2. Near-estimability

Let $\mathbf{X}_1$ be $n \times (p-1)$ of full column rank, $\mathbf{c} \in \mathbb{R}^{p-1}$, and $\mathbf{z}$ a unit vector orthogonal to $\mathcal{C}(\mathbf{X}_1)$. For $\delta > 0$ let $\mathbf{X}_\delta = [\mathbf{X}_1, \mathbf{X}_1\mathbf{c} + \delta\mathbf{z}]$, with least squares estimate $\hat{\boldsymbol{\beta}}_\delta = (\hat{\beta}_1^\top, \hat{\beta}_2^\top)^\top$, and let $\mathbf{X}_0 = [\mathbf{X}_1, \mathbf{X}_1\mathbf{c}]$, which has rank $p-1$. Write $\mathbf{a} = (\mathbf{a}_1^\top, a_2^\top)^\top$.

- (a) $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}_\delta) = \sigma^2 \mathbf{a}_1^\top (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{a}_1 + \sigma^2 (a_2 - \mathbf{c}^\top \mathbf{a}_1)^2 / \delta^2$.
- (b) $\mathbf{a}^\top \boldsymbol{\beta}$ is estimable in the model with matrix $\mathbf{X}_0$ iff $a_2 = \mathbf{c}^\top \mathbf{a}_1$. For such $\mathbf{a}$, $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}_\delta)$ does not depend on δ and equals the variance of the least squares estimator of $\mathbf{a}^\top \boldsymbol{\beta}$ in the limiting model. For every other $\mathbf{a}$ it grows like δ^{-2} as $\delta \rightarrow 0$.

Proof.

- (a) Apply [Theorem 6.6.2](#) with $\mathbf{X}_2 = \mathbf{X}_1$ and the last column in the role of $\mathbf{X}_1$ there. The last column residualized on $\mathcal{C}(\mathbf{X}_1)$ is $(\mathbf{I} - \mathbf{M}_1)(\mathbf{X}_1\mathbf{c} + \delta\mathbf{z}) = \delta\mathbf{z}$, so $\hat{\beta}_2 = \mathbf{z}^\top \mathbf{Y} / \delta$. By part (c) of the same theorem and $\mathbf{X}_1^\top \mathbf{z} = \mathbf{0}$,

$$\hat{\beta}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top (\mathbf{Y} - (\mathbf{X}_1\mathbf{c} + \delta\mathbf{z})\hat{\beta}_2) = \hat{\boldsymbol{\beta}}_1^S - \mathbf{c} \hat{\beta}_2, \quad \hat{\beta}_1^S = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{Y}.$$

Hence $\mathbf{a}^\top \hat{\boldsymbol{\beta}}_\delta = \mathbf{a}_1^\top \hat{\boldsymbol{\beta}}_1^S + (a_2 - \mathbf{c}^\top \mathbf{a}_1) \hat{\beta}_2$. The two terms are uncorrelated, because $\text{Cov}(\hat{\boldsymbol{\beta}}_1^S, \hat{\beta}_2) = \sigma^2 (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{z} / \delta = \mathbf{0}$, and $\text{Var}(\hat{\beta}_2) = \sigma^2 / \delta^2$.

- (b) By Theorem 8.3.1, $\mathbf{a}^\top \boldsymbol{\beta}$ is estimable under $\mathbf{X}_0$ iff $\mathbf{a} \in \mathcal{C}(\mathbf{X}_0^\top)$. Since $\mathbf{X}_1$ has full column rank, $\mathcal{C}(\mathbf{X}_1^\top) = \mathbb{R}^{p-1}$, so $\mathcal{C}(\mathbf{X}_0^\top) = \{(\mathbf{a}_1^\top, \mathbf{c}^\top \mathbf{a}_1)^\top : \mathbf{a}_1 \in \mathbb{R}^{p-1}\}$. For such $\mathbf{a}$ the second term in (a) vanishes. In the limiting model $\mathbf{X}_0 \boldsymbol{\beta} = \mathbf{X}_1 (\boldsymbol{\beta}_1 + \mathbf{c} \beta_2)$, so $\mathbf{a}^\top \boldsymbol{\beta} = \mathbf{a}_1^\top (\boldsymbol{\beta}_1 + \mathbf{c} \beta_2)$ is estimated by $\mathbf{a}_1^\top \hat{\boldsymbol{\beta}}_1^S$, with variance $\sigma^2 \mathbf{a}_1^\top (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{a}_1$. If $a_2 \neq \mathbf{c}^\top \mathbf{a}_1$, the second term is a positive multiple of δ^{-2} . ■

The orthogonality of $\mathbf{z}$ to $\mathcal{C}(\mathbf{X}_1)$ matters only for the size of the bounded term. A general perturbation $\mathbf{w}$ splits as $\mathbf{M}_1 \mathbf{w} + (\mathbf{I} - \mathbf{M}_1) \mathbf{w}$, and the first part, being $\mathbf{X}_1 \mathbf{d}$ for some $\mathbf{d}$, can be absorbed into $\mathbf{c}$. The dichotomy survives: a function estimable in the limit keeps a variance that does not depend on δ , and every other function's variance grows like δ^{-2} . But the bounded value can exceed the limiting-model variance (Exercise (B2)). Figure 8.3.1 showed the δ^{-2} growth numerically; the proposition identifies the rate and the exact set of functions that escape it.

26.1.4 Statistical, not numerical

Chapter 10 met ill-conditioning as a numerical problem: a large $\kappa(\mathbf{X})$ amplifies rounding errors (Theorem 10.5.1). The statistical cost is different. It is σ^2 / λ_ℓ along each weak direction, so it involves the error variance and the absolute size of the eigenvalues, not only their ratio. Rescaling a column changes $\kappa(\mathbf{X})$ (Exercise (B2)) but no estimand's variance, so diagnostics must be computed for a stated scaling (Section 26.3) or be scale-free (Section 26.2). And the eigenvalues grow with n : if the rows are drawn independently from a distribution with nonsingular second-moment matrix $\boldsymbol{\Sigma}$, then $\lambda_p / n \rightarrow \lambda_{\min}(\boldsymbol{\Sigma})$ by the strong law of large numbers and Proposition 10.4.2, so collinearity in the population is paid for by a larger n . Longley's regression shows how far apart the two problems can be: QR computes its coefficients to about eleven digits (Example (Longley's data, exactly)), yet its variance inflation factors reach 1788.5 (Example (Longley's regression)).

Measurement error in the regressors sits between the two. It perturbs $\mathbf{X}$ as rounding does, by an amount that more digits cannot reduce, and a perturbation comparable to $\sqrt{\lambda_p}$ can turn the weak eigenvector and change which combinations appear well estimated (Theorem 10.5.1, with $\kappa(\mathbf{X})$ as the amplifier). The directions that collinearity leaves ill-determined are thus the least robust to errors in $\mathbf{X}$; Chapter 24 treats the resulting bias (Theorem 24.1.1).

26.1.5 Exercises

26.1.5.1 A. Check your understanding

Exercise. A1

In Example (Two regressors that move together), the difference $\beta_1 - \beta_2$ corresponds to $\mathbf{a} = (0, 1, -1)^\top$, of squared length 2. Check that the variance quoted for $\hat{\beta}_1 - \hat{\beta}_2$ lies between the two bounds of Theorem 26.1.1(b), applied to the slope block, and explain why it is close to the upper one.

Solution. Computed without rounding, the variance is $11.640\sigma^2$, and the bounds are $2\sigma^2/70.05 = 0.0286\sigma^2$ and $2\sigma^2/\lambda_2 = 11.643\sigma^2$. (The check needs more digits than the table gives: 3.412^2 and $2/0.1718$ agree to four figures.) The variance is $\sigma^2 \sum_{\ell} (\mathbf{a}^\top \mathbf{v}_\ell)^2 / \lambda_\ell$, and $(\mathbf{a}^\top \mathbf{v}_2)^2 = 1.9995$ of the total $\|\mathbf{a}\|^2 = 2$, because $\mathbf{a}$ points almost exactly along the weak eigenvector. Nearly all of its weight sits on the term with the small eigenvalue.

Exercise. A2

Under normal errors, $\hat{\beta}_2 \sim N(\beta_2, \text{Var}(\hat{\beta}_2))$. With $\beta_2 = 1$ and the standard deviation of [Example \(Two regressors that move together\)](#), compute $\Pr(\hat{\beta}_2 < 0)$ and compare it with the share of replicates in which $\hat{\beta}_2$ was negative.

Exercise. A3

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2]$ have orthogonal columns with $\|\mathbf{x}_1\| = 1$ and $\|\mathbf{x}_2\| = 10$. Find the canonical coordinates and the weak direction. Now measure the second regressor in units ten times larger, so that the column becomes $\mathbf{x}_2/10$. What happens to the weak direction and to $\kappa(\mathbf{X})$? Which variances of estimable quantities have changed?

26.1.5.2 B. Practice

Exercise. B1

Show that $\max_j \text{Var}(\hat{\beta}_j) \geq \sigma^2 / (p\lambda_p)$, so that a small λ_p always hurts at least one coefficient. Give an example with $p = 3$ in which λ_p is tiny but only one coefficient has a large variance.

Solution. By [Theorem 26.1.1\(d\)](#), $\sum_j \text{Var}(\hat{\beta}_j) = \sigma^2 \sum_{\ell} 1/\lambda_{\ell} \geq \sigma^2/\lambda_p$, and the largest of p numbers is at least their average. For the example take $\mathbf{X}$ with orthonormal columns and multiply the third column by ϵ . Then $\mathbf{X}^\top \mathbf{X} = \text{diag}(1, 1, \epsilon^2)$, $\lambda_p = \epsilon^2$, and only $\text{Var}(\hat{\beta}_3) = \sigma^2/\epsilon^2$ is large: a short column, not a near-dependence, which is why [Section 26.3](#) scales the columns.

Exercise. B2

In [Proposition 26.1.2](#), replace $\mathbf{z}$ by an arbitrary vector $\mathbf{w} \notin \mathcal{C}(\mathbf{X}_1)$. Write $\mathbf{w} = \mathbf{X}_1 \mathbf{d} + \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\| \mathbf{z}$ with $\mathbf{z}$ a unit vector orthogonal to $\mathcal{C}(\mathbf{X}_1)$, and show that

$$\text{Var}(\mathbf{a}^\top \hat{\beta}_\delta) = \sigma^2 \mathbf{a}_1^\top (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{a}_1 + \sigma^2 \frac{(a_2 - (\mathbf{c} + \delta \mathbf{d})^\top \mathbf{a}_1)^2}{\delta^2 \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\|^2}.$$

Deduce that if $a_2 = \mathbf{c}^\top \mathbf{a}_1$ the variance does not depend on δ and equals the limiting-model variance plus $\sigma^2 (\mathbf{d}^\top \mathbf{a}_1)^2 / \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\|^2$, which vanishes iff $\mathbf{d}^\top \mathbf{a}_1 = 0$, and that for every other $\mathbf{a}$ it still grows like δ^{-2} .

Solution. $\mathbf{X}_1\mathbf{c} + \delta\mathbf{w} = \mathbf{X}_1(\mathbf{c} + \delta\mathbf{d}) + \delta\|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\| \mathbf{z}$, which is the setting of the proposition with $\mathbf{c}$ replaced by $\mathbf{c} + \delta\mathbf{d}$ and δ by $\delta\|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\|$. Part (a) gives the formula. If $a_2 = \mathbf{c}^\top \mathbf{a}_1$, the numerator of the second term is $\delta^2(\mathbf{d}^\top \mathbf{a}_1)^2$, the δ^2 cancels, and the second term is the constant $\sigma^2(\mathbf{d}^\top \mathbf{a}_1)^2 / \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\|^2$; the first is the limiting-model variance, by part (b) of the proposition. Otherwise the numerator tends to $(a_2 - \mathbf{c}^\top \mathbf{a}_1)^2 > 0$ and the term grows like δ^{-2} .

Exercise. B3

Section 8.3 used twelve equally spaced t_i on $[-1, 1]$, $\mathbf{X}_1 = [\mathbf{1}, \mathbf{t}]$, $\mathbf{c} = (0, 1)^\top$ and $\mathbf{w} = (\cos \pi t_i)$, and reported $\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = 0.212 \sigma^2$ whatever δ . Use Exercise (B2) to explain the number: show that $\mathbf{d} = (d_1, 0)^\top$ because $\mathbf{w}$ is even and $\mathbf{t}$ is odd, and that the variance equals $\sigma^2 / \sum_i t_i^2$.

Solution. Here $\mathbf{a} = (0, 1, 1)^\top$, so $\mathbf{a}_1 = (0, 1)^\top$ and $a_2 = \mathbf{c}^\top \mathbf{a}_1 = 1$: the function is estimable in the limit. The coefficients $\mathbf{d}$ of the projection of $\mathbf{w}$ on $\mathcal{C}(\mathbf{1}, \mathbf{t})$ are $d_1 = \bar{w}$ and $d_2 = \mathbf{t}^\top \mathbf{w} / \mathbf{t}^\top \mathbf{t} = 0$, because the t_i are symmetric about zero and $\cos$ is even. So $\mathbf{d}^\top \mathbf{a}_1 = d_2 = 0$ and the second term vanishes. The first is $\sigma^2[(\mathbf{X}_1^\top \mathbf{X}_1)^{-1}]_{22} = \sigma^2 / \sum_i t_i^2$, since $\mathbf{1} \perp \mathbf{t}$. With $t_i = -1 + 2i/11$, $i = 0, \dots, 11$, $\sum_i t_i^2 = (4/121) \sum_{i=0}^{11} (i - 5.5)^2 = (4/121) \cdot 143 = 4.727$, and $1/4.727 = 0.2115$, which rounds to 0.212.

26.1.5.3 C. Going deeper

Exercise. C1

A near-dependence need not show up in any pair of columns. Let $\mathbf{X}$ have $p \geq 3$ centred columns of unit length with all pairwise correlations equal to ρ , so that $\mathbf{X}^\top \mathbf{X} = \mathbf{R} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$.

- Show that the eigenvalues of $\mathbf{R}$ are $1 + (p - 1)\rho$, with eigenvector $\mathbf{1}$, and $1 - \rho$ with multiplicity $p - 1$; hence $\mathbf{R}$ is positive definite exactly when $-1/(p - 1) < \rho < 1$.
- Show that at $\rho = -1/(p - 1)$ the columns satisfy $\mathbf{X}\mathbf{1} = \mathbf{0}$ exactly. With $p = 10$ every pairwise correlation is then $-\frac{1}{9}$ in absolute value, a number no inspection of the correlation matrix would remark on, and the design is singular.
- Put $\rho = -1/(p - 1) + \epsilon$ with $\epsilon > 0$ small. Find λ_p , find $\text{Var}(\mathbf{1}^\top \hat{\boldsymbol{\beta}})$, and show that both are governed by ϵ alone.
- Verify that

$$\mathbf{R}^{-1} = \frac{1}{1 - \rho} \left\{ \mathbf{I} - \frac{\rho}{1 + (p - 1)\rho} \mathbf{1}\mathbf{1}^\top \right\},$$

so that $\text{Var}(\hat{\beta}_j)/\sigma^2 = (\mathbf{R}^{-1})_{jj}$, the variance inflation factor of Definition 26.2.1, is $(1 - \rho)^{-1} \{1 - \rho/(1 + (p - 1)\rho)\}$. Evaluate it at $p = 10$ and $\rho = -\frac{1}{9} + \epsilon$ and show that it grows like $1/(90\epsilon)$.

What does the exercise say about diagnosing collinearity from a table of correlations?

Solution.

- (a) $\mathbf{R}\mathbf{1} = (1 - \rho)\mathbf{1} + \rho p\mathbf{1} = \{1 + (p - 1)\rho\}\mathbf{1}$; and if $\mathbf{v} \perp \mathbf{1}$ then $\mathbf{R}\mathbf{v} = (1 - \rho)\mathbf{v}$. These account for all p eigenvalues, and $\mathbf{R}$ is positive definite iff both are positive.
- (b) At $\rho = -1/(p - 1)$ the first eigenvalue is $1 - 1 = 0$, so $\mathbf{R}\mathbf{1} = \mathbf{0}$, and $\|\mathbf{X}\mathbf{1}\|^2 = \mathbf{1}^\top \mathbf{X}^\top \mathbf{X}\mathbf{1} = 0$, whence $\mathbf{X}\mathbf{1} = \mathbf{0}$: the columns sum to the zero vector. With $p = 10$, $\rho = -\frac{1}{9} = -0.111$.
- (c) The eigenvalue on $\mathbf{1}$ becomes $1 + (p - 1)\{-1/(p - 1) + \epsilon\} = (p - 1)\epsilon$, which for small ϵ is the smallest, so $\lambda_p = (p - 1)\epsilon$ and $\mathbf{v}_p = \mathbf{1}/\sqrt{p}$. By [Theorem 26.1.1\(a\)](#) applied to that eigenvector, $\text{Var}(\mathbf{1}^\top \hat{\boldsymbol{\beta}}) = p \text{Var}(\mathbf{v}_p^\top \hat{\boldsymbol{\beta}}) = p\sigma^2/\{(p - 1)\epsilon\}$, which is also the upper bound of [Theorem 26.1.1\(b\)](#) at $\mathbf{a} = \mathbf{1}$, attained because $\mathbf{1}$ points exactly along the weak eigenvector.
- (d) Multiply out: with $c = \rho/\{1 + (p - 1)\rho\}$, $\{(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\}(\mathbf{I} - c\mathbf{1}\mathbf{1}^\top) = (1 - \rho)\mathbf{I} + \mathbf{1}\mathbf{1}^\top\{\rho - c(1 - \rho + \rho p)\} = (1 - \rho)\mathbf{I}$, since $1 - \rho + \rho p = 1 + (p - 1)\rho$. Hence the stated inverse, and $(\mathbf{R}^{-1})_{jj} = (1 - \rho)^{-1}(1 - c)$. At $p = 10$ and $\rho = -\frac{1}{9} + \epsilon$, $1 + 9\rho = 9\epsilon$ and $1 - \rho = \frac{10}{9} - \epsilon$, so $c = \rho/(9\epsilon) \rightarrow -1/(81\epsilon)$ and $(\mathbf{R}^{-1})_{jj} \approx \frac{9}{10}\{1 + 1/(81\epsilon)\} = \frac{9}{10} + 1/(90\epsilon)$.

A table of pairwise correlations answers the question “is any one column nearly a multiple of another?” The question that matters is “is any combination of the columns nearly zero?”, and the two are different as soon as $p > 2$: here ten columns that are nearly mutually orthogonal in pairs are nevertheless nearly linearly dependent as a set. The eigenvalues of [Theorem 26.1.1](#) and the variance inflation factors see it; the correlation matrix, read entry by entry, does not.

Exercise. C2

Collinearity does not merely spread the same total precision differently: it can put *more* precision on a chosen direction than an orthogonal design of the same column lengths.

- (a) Let $p = 2$ with centred columns of length L and correlation $\rho \in (-1, 1)$. Show that

$$\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = \frac{2\sigma^2}{L^2(1 + \rho)},$$

which decreases as ρ increases, and is half its orthogonal value as $\rho \rightarrow 1$.

- (b) For p equicorrelated columns of length L , show that $\text{Var}(\mathbf{1}^\top \hat{\boldsymbol{\beta}}) = p\sigma^2/[L^2\{1 + (p - 1)\rho\}]$, so that the sum of the coefficients is estimated p times better than in an orthogonal design as $\rho \rightarrow 1$, and infinitely worse as $\rho \rightarrow -1/(p - 1)$, the design of [Exercise \(C1\)](#).
- (c) Reconcile this with [Theorem 26.1.1\(b\)](#) and (d).

Solution.

- (a) $\mathbf{X}^\top \mathbf{X} = L^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, whose inverse is $\{L^2(1 - \rho^2)\}^{-1} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$. With $\mathbf{a} = (1, 1)^\top$, $\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a} = (2 - 2\rho)/\{L^2(1 - \rho^2)\} = 2/\{L^2(1 + \rho)\}$. At $\rho = 0$ this is $2/L^2$ and at $\rho \rightarrow 1$ it is $1/L^2$.
- (b) $\mathbf{X}^\top \mathbf{X} = L^2 \mathbf{R}$ with $\mathbf{R}$ as in [Exercise \(C1\)](#), and $\mathbf{R}^{-1}\mathbf{1} = \mathbf{1}/\{1 + (p - 1)\rho\}$ because $\mathbf{1}$ is an eigenvector. So $\mathbf{1}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{1} = p/[L^2\{1 + (p - 1)\rho\}]$. At $\rho = 0$ this is p/L^2 ; as $\rho \rightarrow 1$ it is $1/L^2$, smaller by the factor p ; as $\rho \rightarrow -1/(p - 1)$ it diverges.
- (c) There is no contradiction. Part (b) of the theorem bounds $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})$ by $\sigma^2 \|\mathbf{a}\|^2 / \lambda_p$

and below by $\sigma^2 \|\mathbf{a}\|^2 / \lambda_1$, and here $\mathbf{a} = \mathbf{1}$ has $\|\mathbf{a}\|^2 = p$ and points exactly along the *strong* eigenvector when $\rho > 0$, so the lower bound $p\sigma^2 / \lambda_1$ is attained: collinearity has made $\lambda_1 = L^2\{1 + (p-1)\rho\}$ as large as it can be, and the sum is the direction that benefits. Part (d) records the price: $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$ contains the $p-1$ terms $\sigma^2 / \{L^2(1-\rho)\}$, which diverge as $\rho \rightarrow 1$. Total precision is not conserved either; what is true is that the design decides how much of it each direction receives, and a design chosen with one estimand in mind can be better than an orthogonal one for that estimand and far worse for every other.